

ROVER LAB, DOSSIER (TRAPPIST-1 e mission)

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Planet

Name: TRAPPIST-1 e

Planet type: Temperate rocky planet, one of seven worlds orbiting an ultracool red dwarf

Surface temperature: ≈ -24 °C equilibrium estimate (range reported -48 to -23 °C depending on assumed albedo). Tidally locked.

Surface pressure: Not directly measured; no atmosphere confirmed. 2025 JWST observations were inconclusive.

Gravity: ≈ 0.82 g

Atmosphere: Not confirmed. Data are consistent with either a bare rock or a thin nitrogen-dominated atmosphere (hypothesis, not measurement).

Mission Briefing

TRAPPIST-1 e is a rocky, Earth-sized planet in the habitable zone of an ultracool red dwarf star, tidally locked so that one face permanently points toward its sun. The rover 'Terra-e' has landed near the twilight terminator, the only region with a stable, moderate climate between the scorched dayside and the frozen nightside. Orbital spectrometry has flagged a rust-orange basin nearby, likely a dry lakebed where mineral-rich water once pooled and evaporated. Given that TRAPPIST-1 planets are unusually dense for their size, hinting at large iron cores, mission control wants to know whether this basin holds a genuinely iron-rich brine that could be processed into iron oxide for the station's radiation shielding.

Rover

Starting energy: 100

Site visit cost: 10

Experiment cost: 15

Overview Image

Site: Rust-orange basin near the terminator

Description: A shallow, dry basin streaked with rust-orange mineral crusts, ringed by darker rock at the edge of the permanent twilight zone.

Identity: Iron(III) chloride solution (real sample, suggested ≈ 0.4 M), medium-high concentration. REAL SAMPLE GROUP.

Known Properties

Colour: Amber-orange

Odour: Faint, sharp

Solubility: Fully miscible with water

Experiment Results

Measure pH: Strongly acidic, pH around 1.4–1.7

Measure density: 1.25 g/mL, denser than water

Test electrical conductivity: High

Add a magnet: No attraction

Add potassium thiocyanate solution: Turns deep blood-red almost immediately

Add sodium hydroxide solution: Forms a substantial reddish-brown gelatinous precipitate

Heat a small sample gently: Darkens slightly

Experiments

Measure pH

Measure density

Test electrical conductivity

Add a magnet

Add potassium thiocyanate solution

Add sodium hydroxide solution

Heat a small sample gently

AI Behaviour

Reveal remaining candidates after narrowing: No

Socratic questions: Yes

Difficulty: Standard (15-16 years old)

Teacher Notes

REAL SAMPLE GROUP, assign real FeCl₃, medium-high concentration (suggested ≈ 0.4 M). TRAPPIST-1e is temperate (cooler/less evaporative than 55 Cancri e), so it's positioned as the second-most-concentrated of the five real groups.

Real-science notes: TRAPPIST-1e is a genuine rocky exoplanet, ~ 40.7 light-years away, discovered in 2017 (Spitzer + TRAPPIST telescope, Gillon et al.), orbiting an M8 ultracool dwarf star every 6.1 days. Agol et al. (2021) found that TRAPPIST-1 planets as a group are unusually dense for their size, consistent with unusually large iron cores, TRAPPIST-1e itself is actually one of the least dense (most Earth-like) planets in the system, close to Earth's own bulk density (~ 4.9 g/cm³ vs Earth's 5.51 g/cm³). No atmosphere or surface chemistry has actually been detected on TRAPPIST-1e, 2025 JWST data were inconclusive. TRAPPIST-1 is a genuinely active flaring star, which matters for the biology lesson (see below): frequent stellar flares bathe the whole system in periodic bursts of UV/X-ray radiation.

Sources: Agol et al. 2021 and Gillon et al. 2017 (see en.wikipedia.org/wiki/TRAPPIST-1e for citations); NASA Exoplanet Archive (exoplanetarchive.ipac.caltech.edu/overview/TRAPPIST-1).

Cross-subject link: Biology selective pressure for this planet, intense stellar flares and radiation from an active red dwarf star. Pairs naturally with TRAPPIST-1 g (same system, cooler orbit) for a cross-group comparison at the final conference.